Predicting ΔΔG of protein–protein binding from sequence: how much of a fine-tuned ESM-2's accuracy is leakage?

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Fine-tuning ESM-2 to predict the binding cost of a single interface mutation reaches **Spearman ρ = 0.66** when the test set is drawn at the mutation level, and **ρ = 0.18** when no protein complex is allowed to appear in both training and test. On the honest split, 148M fine-tuned parameters are **statistically indistinguishable** from a frozen backbone with a 0.49M-parameter head (paired over 5 replicates, p = 0.23). The 3.7× gap between the two split definitions is the result this report is about.

1. Problem — and why it is a reasonable proxy for antibody optimization

ΔΔG of binding is the quantity an affinity-maturation campaign searches over: $\Delta\Delta G = \Delta G_{mut} - \Delta G_{WT}$, with $\Delta G = RT \cdot \ln K_d$, so positive means the mutation weakens the interface. Ranking candidate interface substitutions by predicted ΔΔG is the primitive a binder-engineering loop needs before committing to expression and SPR: it need not be calibrated in kcal/mol, it has to put the right mutations at the top of the list — which is why **Spearman is the primary metric here, with RMSE reported alongside rather than instead**. The sequence-only formulation is the deliberately harder and more useful one, since it applies to the many antibody–antigen pairs with no solved co-crystal, and SKEMPI 2.0 makes it measurable: experimental K_d for wild-type and mutant complexes of known structure.

TO WRITE — THE LINK TO YOUR OWN INTERFACE-ENERGETICS WORK

Two or three sentences on what your published work established experimentally, and why a sequence-only learned model is the complementary question rather than a restatement of it. Left blank deliberately.

2. Data, and the split that decides everything

Dataset. SKEMPI 2.0 ships 7,085 mutation records; restricting to single-point mutations leaves 5,112. A further **127 rows are dropped because their affinity is a bound, not a measurement** — SKEMPI's convenience columns `Affinity_*_parsed` strip the leading < or > and return a bare number, so filtering on missing values alone silently keeps "no detectable binding" entries as if they were real K_d. The filter runs against the raw affinity columns, where the marker survives. Dropping 156 further rows with unusable K_d gives **4,829 mutations across 316 complexes**. Sequences are reconstructed per complex from the residue-mapping files SKEMPI ships with each structure; the wild-type residue named by each mutation code matches the reconstruction for **5,112 / 5,112 rows (100%)**, asserted at parse time because an off-by-one here is what this pipeline is most exposed to. A ≤2,048-token length policy then excludes two large multi-chain assemblies, leaving **4,814 mutations across 314 complexes**. Labels skew hard toward destabilization (3,612 positive vs 1,050 negative; mean +0.96, median +0.56, sd 1.71 kcal/mol) with **11.0% of rows beyond |ΔΔG| > 3** — hence Huber loss rather than MSE.

The split. This is what determines whether any number below means anything. Mutations of the same complex are highly correlated — SKEMPI contains dozens of alanine scans of the same interface. Under a random row-level partition, nearly every complex in the test set also appears in training, so the model can memorise a complex from its training mutations and then "predict" held-out mutations of that same interface. Memorising and learning ΔΔG become indistinguishable, and both register as progress. **This project therefore partitions groups, never rows**, and reports three definitions side by side rather than choosing one:

Split definition	Groups	What it holds out
`split_mutation`	—	Naive row-level partition. **Leaks by construction.** Reported as the contrast case, never the headline.
`split_pdb_id`	313	Whole structures. The literal reading of "no complex in two splits".
`split_hold_out_proteins`	152	SKEMPI's curated protein-pair grouping. Strictest — also catches one protein pair appearing under several PDB entries.

Assignment is size-aware rather than a uniform sample of groups: SKEMPI's group sizes span three orders of magnitude, and sampling uniformly produced a 4.7% validation set where 15% was intended — too thin to select a model on. Groups are instead shuffled under a fixed seed, taken largest-first, and each given to whichever split is furthest below its row target, landing all three definitions at 70.0 / 15.0 / 15.0 by row count with groups kept atomic. The invariant is therefore structural rather than merely tested, and it is also tested: 14 `pytest` cases assert that no group crosses a split boundary. **The cost of size-aware assignment, stated rather than buried:** filling `train` first sends the large, well-studied complexes there (121 training complexes vs ~96 each in val and test), so val and test hold smaller, less-studied ones — arguably the realistic generalization target, but a structured train/test difference rather than a purely random one.

3. Model and setup

A **siamese ESM-2**: one shared backbone encodes the wild-type and the mutant complex. Weight sharing is the point — it puts both embeddings in one space, so their difference is meaningful. Each is mask-aware mean-pooled, and the head sees `concat[wt, mut, mut - wt]`, since ΔΔG is a question about what changed. Both partner chains are concatenated into one sequence, so the backbone sees both sides of the interface at once.

BACKBONE
`facebook/esm2_t30_150M_UR50D` — 30 layers, hidden 640

BATCHING
By **token budget** — 4,096 padded tokens, not a fixed example count

HEAD
Linear(1920→256) → ReLU → Dropout(0.1) → Linear(256→1)

TRAINABLE PARAMETERS
148.6M, full fine-tune — of which the head is 0.49M

LOSS / OPTIMIZER
Huber δ = 1.0; AdamW, lr 2×10^{-5}, no schedule

PRECISION
bf16 autocast, no GradScaler (bf16 keeps fp32's exponent range)

EPOCHS / SELECTION
≤10, early stopping patience 3; best **validation Spearman**, not val loss

HARDWARE
1× RTX 4090 24 GB, Windows 11, PyTorch 2.13 / CUDA 13.0

WALL CLOCK
179 s/epoch, ≈25 min per run; 6.2 GPU-hours for all 15 replicate runs

REPLICATES
5 per method per split — 3 training seeds at fixed partition, 3 split seeds at fixed training seed

Full fine-tuning rather than LoRA, and 150M rather than 650M — both measured, not assumed. 150M full peaks at 10.9 GB; 650M needs 26.9 GB on a 24 GB card even with gradient checkpointing, because the siamese design calls the backbone twice per step, and runs **33× slower** once traffic spills to host memory — silently, since CUDA on Windows falls back to shared system memory rather than raising `OutOfMemoryError`. Two baselines keep the fine-tuned model from failing silently: zero-shot masked-marginal scoring fixes the floor, a frozen backbone with a trained head measures what the pretrained representation already knows. And because the three split definitions disagree about which rows are held out, **a separate model is trained per definition**, with the harness marking the other two contaminated.

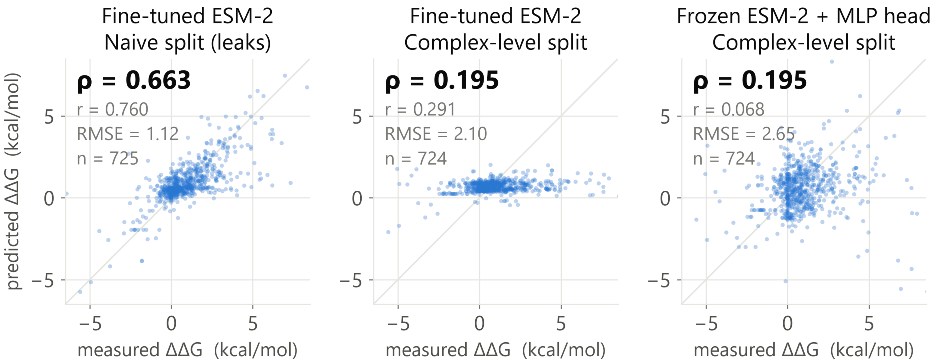

Figure 1 — the same model, the same data, two split definitions. Predicted vs measured ΔΔG on held-out test rows; the diagonal is $y = x$. **Left:** the naive split, where the same complexes appear in training. **Centre:** identical model and hyperparameters under a complex-level split — the prediction range collapses toward the training mean, which is what ρ = 0.20 looks like. **Right:** a frozen backbone with a 0.49M head on the same partition, matching it. Panels show the **median** of five replicates, not the best; the probe is drawn on the same partition as the fine-tuned model beside it.

4. Results — test Spearman ρ, mean ± sd over 5 replicates

Model	Trainable params	`split_mutation` naive — leaks	`split_pdb_id` honest	`split_hold_out_proteins` honest, strictest
Zero-shot ESM-2 150M (masked-marginal)	0	−0.082	+0.005	−0.097
Frozen ESM-2 + ridge	1.9K	0.473 ± 0.021	0.053 ± 0.057	0.122 ± 0.102
Frozen ESM-2 + MLP head	0.49M	0.623 ± 0.024	**0.223 ± 0.020**	**0.272 ± 0.053**
Fine-tuned ESM-2 150M	148.6M	**0.660 ± 0.021**	0.178 ± 0.059	0.167 ± 0.091

Zero-shot is a single deterministic, ranking-only run, so it has no error bars — but its flatness is the control that matters: **the three test sets are of similar intrinsic difficulty**, so the gaps across the columns come from training, not from one split being easier. That floor is explained rather than merely reported: scoring the mutated chain *alone* and inside the full complex agree at ρ = 0.888, so ESM-2's masked-marginal is blind to the binding partner — it scores whether a residue belongs in its chain, not whether it holds an interface together.

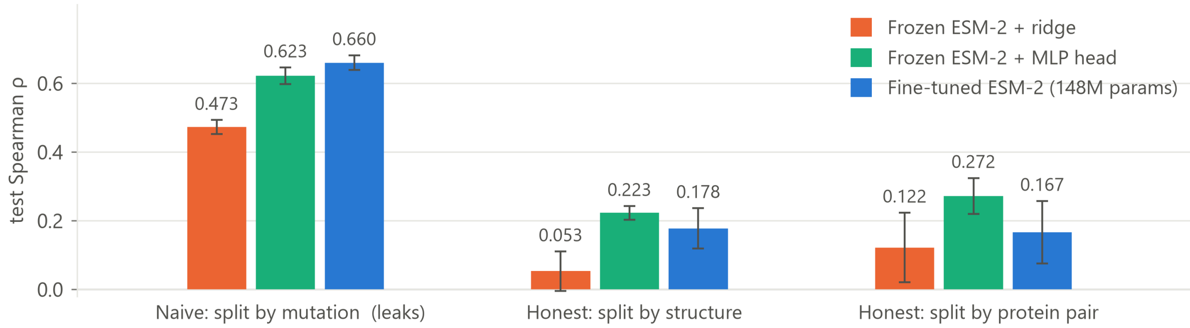

Figure 2 — the leakage gradient. Test Spearman by method under each split definition; bars are the mean of 5 replicates, whiskers ±1 sd. Note the *within-group* order too: on the leaky split more trainable parameters help monotonically; on both honest splits that ordering disappears.

Fine-tuning wins on exactly one split — the leaky one

Replicates are paired on identical partitions, so this compares fine-tuning against the frozen MLP probe on the same data rather than as two independent samples:

Split definition	Mean Δρ (fine-tune − probe)	95% CI	p	Verdict
`split_mutation` (leaks)	**+0.038**	[+0.003, +0.072]	**0.038**	Fine-tuning genuinely better
`split_pdb_id`	−0.045	[−0.133, +0.043]	0.229	Not distinguishable
`split_hold_out_proteins`	−0.105	[−0.256, +0.045]	0.124	Not distinguishable

The only split on which 148M trainable parameters demonstrably beat a frozen backbone plus a 0.49M head is the one that rewards memorisation. The training curves say why, across all five replicates rather than in one illustrative run. Under the naive split, validation holds other mutations of complexes already in training, so the best epoch is the **last** one in all five runs (val ρ 0.17–0.35 at epoch 0 rising to 0.62–0.68 at epoch 9) — it never signals a stop. Under `split_pdb_id` the best epoch is erratic: 3, 3, 7, 9, 9. Under the strictest split, **three of five replicates peak at epoch 0**, before the model has learned anything transferable from the training complexes at all. Same model, same hyperparameters, same data — the only variable is whether the split rewards memorising a complex. The diagnosis is ordinary overfitting, not a bug: 148M parameters against 3,374 training rows from 121 complexes, ending at train $\rho = 0.87$ against test $\rho = 0.22$.

A second finding worth as much as the first: fine-tuning is 2–4× more variable than the probe (sd 0.048–0.113 vs 0.016–0.063), and fixing the random seed does not remove it — the identical configuration produced 0.148 and 0.225 on two different days, from CUDA kernel nondeterminism alone. A single fine-tuning run of this model is not a reportable number, which is why everything above is a 5-replicate mean.

The antibody–antigen subset

SKEMPI labels antibody–antigen complexes, so this breakdown is free — and it is the subset that matters most for binder engineering. It was pre-specified before any results existed, making it a planned comparison rather than a subgroup found by searching.

AB/AG test rows only (189–230 rows)	`split_pdb_id` — honest	`split_mutation` — leaks
Frozen ESM-2 + MLP head	**+0.264 ± 0.052**	+0.551 ± 0.042
Fine-tuned ESM-2 150M	**−0.065 ± 0.142**	+0.497 ± 0.083
Paired difference (fine-tune − probe)	**−0.330**, [−0.540, −0.119], $p = 0.012$	−0.054, [−0.121, +0.012], $p = 0.086$

Across all test rows, fine-tuning and the frozen probe are indistinguishable. **On antibody complexes they are not: fine-tuning is reliably worse, and its Spearman is below zero** — the ranking is anti-correlated with measured ΔΔG, in all five replicate pairs. Two caveats belong with it: the strictest split holds only 6 AB/AG test rows against a 10-row minimum, so the harness omits the metric there and the claim rests on `split_pdb_id` alone; and it is one subgroup among several in the error analysis — a pre-registered signal, not a confirmed mechanism.

5. Limitations

- **The honest numbers are modest ($\rho \approx 0.17$–0.27), and they are the reported result.** A correctly attributed number on an honest split is worth more than an inflated one; the 0.66 is included to show what a split choice is worth.
- **Sequence-only**, although SKEMPI ships structures, and **single-point only**, dropping 1,973 multi-point rows, so nothing here speaks to epistasis. No comparison against FoldX-style physics-based methods on their own terms is attempted.
- **Mean-pooling discards position** — a whole-sequence mean is a weak signal for a change of one residue in up to 2,048, plausibly the largest architectural limitation here. Separately, **127 inequality rows are dropped rather than used** (censored regression would use them properly), and val/test hold smaller, less-studied complexes as a side effect of size-aware assignment (§2).
- **Rows between 1,024 and 2,048 tokens (1.6%) exceed ESM-2's pretraining crop;** rotary embeddings extrapolate, but quality there is not guaranteed. Truncation was rejected because at a 1,024 cap the mutation site falls outside the window for 77 rows, making `mut - wt` exactly zero while the model still trains against a real label — a silent failure rather than a loud one.

6. Next steps — the model overfits and is unstable, so the goal is to cut variance, not only to raise the mean

- **Position-specific pooling at the mutated residue,** concatenated with the mean-pooled context; the dataset already carries the token index for it. Targets the limitation the honest splits expose most directly, and is the highest-value single change.
- **LoRA on the 150M backbone as regularisation** — a small adapter should shrink the replicate spread that makes single runs unreportable, and it doubles as the prerequisite for 650M, which does not fit on this card under full fine-tuning. Alongside it, a lower learning rate or freezing the lower N layers, aimed at the epoch-0 peak on the strictest split.
- **Diagnose the antibody-subset inversion** before trusting any antibody number: is CDR sequence variability being read as noise, or are the loops simply out of distribution for a general protein language model? Then, and only then, structural features — relative solvent accessibility and interface contact counts are cheap given SKEMPI's structures, and would test whether the ceiling is the representation or the task.

TO WRITE — RELATED WORK, IN YOUR OWN WORDS, WITH LINKS

One paragraph positioning against MINT, ProBASS, AbTune, Seq2Bind and 3D-ΔΔG — AbTune is the natural comparison, being antibody-focused, which makes §4's AB/AG numbers the place to compare. Worth checking rather than asserting: which of them report on complex-level rather than mutation-level splits.

Data: SKEMPI 2.0 (Jankauskaitė et al., *Bioinformatics* 2019), life.bsc.es/pid/skempi2 — verify redistribution terms before publishing the processed tables. Weights: `facebook/esm2_t30_150M_UR50D`. Every figure and table here regenerates from `results/metrics.json` via `report/make_figures.py` and `python -m src.replicates`.